Expansion Properties of the Superspecial Isogeny Graph with Level Structure

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ENS de Lyon
Isogeny Club, Season 8

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Based on ongoing work with Maria Corte-Real Santos, Olivier Taïbi, and Benjamin Wesolowski

Outline

1. Motivation
2. The supersingular ℓ-isogeny graph with extra data
3. Background
4. Higher dimensional graph
5. Main results
6. The g-dimensional superspecial (ℓ-)isogeny graph with extra data
7. Applications
8. A brief overview of the techniques for the main results

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There are n edges from E_1 to E_2 if there exist n distinct subgroups $G \subset E_1[\ell]$ of order ℓ such that $E_1/G \simeq E_2$.

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3. Finite graph: $|\mathcal{V}| \sim \frac{p}{12}$.

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Ramanujan graph and Crypto...

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- Extend the graph $\mathcal{G}_\ell$: Add finite data (set) to each vertex and request the edges compatible with the extra data.

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Do these rich structural and expansion properties carry over to principally polarized abelian varieties of dimension $g > 1$?

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- Ready to introduce the vertices and edges for the higher dimensional graphs.

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- *containing only the edges of (reduced) degree ℓ.*

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Close look on the edges of $\mathcal{G}_\ell(p, g)$

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$$\frac{|\{\ell\text{-isogenies } X \to Y\}|}{|\mathrm{Aut}(Y)|}.$$

- Let X and Y be two vertices in $\mathcal{V}$. There are n edges from X to Y if there exist n distinct **maximally isotropic subgroups** $G \subset X[\ell]$ such that $X/G \simeq Y$.
- **Fact:** If (X, λ) is a superspecial PPAV of dim g, then

$$N_g(\ell) := |\{\text{maximally isotropic subgroup of } X[\ell]\}| = \prod_{i=1}^{g}(\ell^i + 1).$$

II. The Graphs

Properties of $\mathcal{G}_\ell(p, g)$:

1. Finite graph: $|\mathcal{V}| \sim p^{g(g+1)/2}$ by Hashimoto-Ibukiyama (80), Ekedahl (87).

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Consider the graph: $\mathcal{G}_2(11, 2)$, i.e., $\ell = 2$, $g = 2$, $p = 11$.

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$$A_{\mathcal{G}_2(11,2)} = \begin{bmatrix} 3 & 9 & 0 & 3 & 0 \\ 4 & 3 & 4 & 4 & 0 \\ 0 & 3 & 6 & 0 & 6 \\ 1 & 3 & 0 & 3 & 8 \\ 0 & 0 & 3 & 4 & 8 \end{bmatrix}$$

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Question

How far is the graph from being Ramanujan? Is it an ϵ-**expander**?

The first main result: (First bound)

Definition 5.

A regular n-graph G is called ϵ-**expander** if

$$\lambda(G) \leq (1-\epsilon)n, \quad \text{for some } \epsilon < 1,$$

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Theorem [K., Corte-Real Santos, Taïbi, Wesolowski]

$\mathcal{G}_\ell(p,g)$ is an ϵ-expander with

$$\epsilon = 1 - \left(\frac{2\sqrt{\ell}}{\ell+1}\right)^g.$$

Equivalently, $\lambda(\mathcal{G}_\ell(p,g)) \leq N_g(\ell)\left(\frac{2\sqrt{\ell}}{\ell+1}\right)^g = \prod_{i=1}^{g}(1+\ell^i)\left(\frac{2\sqrt{\ell}}{\ell+1}\right)^g$.

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NOTE: This is not the sharpest bound we have obtained! (The better bound is in the next slides).

Quick Observations

Normalize

$\lambda'(\mathcal{G}_\ell(p,g)) := \frac{\lambda(\mathcal{G}_\ell(p,g))}{\deg(\mathcal{G}_\ell(p,g))}$.

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- Our bound:

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- Our bound:

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- Their bound:

$$\lambda'(\mathcal{G}_\ell(p,g)) \leq 1 - \frac{1}{4N_g(\ell)(g+2)} \cdot \left(\frac{\ell-1}{2(\ell-1)+3\sqrt{2\ell(\ell+1)}}\right)^2.$$

Comparison of Eigenvalue Bounds

- Numerically: The bounds for $\lambda'(\mathcal{G}_\ell(p,g))$:

g	ℓ	**[ATY]**	**Ours**	$N_g(\ell)$
2	2	0.99997	**0.88889**	15
	3	0.99998	**0.75000**	40
	5	0.99999	**0.55556**	156
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2) Numerical Data: When $11 \leq p \leq 601$ and $\ell = g = 2$, Florit and Smith (2021) computed (then conjectured for all p):

$$\frac{11}{15} \leq \lambda'(\mathcal{G}_2(p,2)) \leq \frac{12}{15} = 0.8$$

The second main result: (Sharper bound)

Theorem [K., Corte-Real Santos, Taïbi, Wesolowski]

$\mathcal{G}_\ell(p, g)$ is an ϵ-expander with

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Comparing Our Bounds

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Observation II

While First Bound already establishes rapid mixing, **Sharper Bound** drops dramatically as g and ℓ grow, yielding a stronger spectral gap.

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The graph with torsion points

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Despite the level structures, the spectral gap is completely preserved.

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More general

Our main result applies much more general settings than the ones illustrated in two examples (allowing for extra data of arbitrary nature, only requiring it to satisfy a simple property).

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Give two superspecial PPAVs X and X' of dimension $g > 1$, find an isogeny $\phi : X \to X'$.

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- We provide an algorithm solving Problem 1 with heuristic free:
- The key point: Take long enough paths of ℓ-isogeny steps until the endpoint is product/split/non-Jacobians.
- By the main results, the graph $\mathcal{G}_\ell(p, g)$ is an expander, random walks reach a uniform distribution fast.

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Find a 2-isogeny path in $\mathcal{G}_2(p, g)$:

$$\alpha : X \to E_1 \times \cdots \times E_g$$

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Path to Product: dimension 2

$|\mathcal{V}| \sim p^3$:
$|\{\text{Jacobians}\}| \sim p^3$,
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By the Main result, the target of each path is uniformly distributed: The probability is $> 1/p$.

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3. Pullback

Compute bases $(\alpha_i)_i$ and $(\alpha_i')_i$ for $E_1 \times \cdots \times E_g$ and $E_1' \times \cdots \times E_g'$. Set:

$$\beta_i = \alpha^\dagger \alpha_i \alpha \quad \text{and} \quad \beta_i' = \alpha'^\dagger \alpha_i' \alpha'$$

Adjoint isogeny

For any d-isogeny $f : (X, \lambda) \to (X', \lambda')$, the *adjoint* d-isogeny $f^\dagger := \lambda^{-1} \hat{f} \lambda' : X' \to X$.

Application 2: Computing End(X)

Problem 2:

Compute End(X) of a superspecial PPAV X of dimension $g > 1$.

We modify the idea given for surfaces by Anni, Bisson, Iezzi, Garcia, Wesolowski (2025):
1. By the bases of End(E_i), find bases for Hom(E_i, E_j), so find End($E_1 \times \cdots \times E_g$).

2. Path to Product

Find a 2-isogeny path in $\mathcal{G}_2(p, g)$:

$$\alpha : X \to E_1 \times \cdots \times E_g$$

Find a 3-isogeny path in $\mathcal{G}_3(p, g)$:

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Final Result

$\langle (\beta_i)_i, (\beta_i')_i \rangle$ generates the ring End(X).

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- $\mathcal{G}_\ell(p,g)$ (superspecial graph)
- $\mathcal{G}_\ell^{\mathcal{F}}(p,g)$ with level structure

$\Longleftrightarrow$

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Techniques

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THANK YOU

Thank You!